

Chapter 8: Nature of Matter

Elements, Compounds, and Mixtures | Class 8 Science Notes

1. Introduction & The Concept of Matter

Matter is anything that has mass and occupies space. Everything around us—such as air, water, food, books, trees, and buildings—is composed of matter and made up of tiny particles. However, non-physical phenomena such as light, heat, electricity, thoughts, and emotions are **not matter**.

Scientific vs. Common Meaning of 'Pure'

In common usage: 'Pure' means unadulterated products (e.g., pure milk, ghee, spices).

In science: A *pure substance* consists of only one single type of particle throughout and cannot be separated into other substances by physical processes.

2. What Are Mixtures?

A **mixture** is formed when two or more substances are combined physically without reacting chemically. The individual constituent substances are called **components**, and each component retains its original properties.

Type of Mixture	Characteristics	Examples from Chapter
Non-Uniform (Heterogeneous)	Components are visually distinguishable by naked eye or magnifying device; non-evenly distributed.	Sprout salad (chickpeas, onion, tomato), poha, sand in water, oil & water, muddy water.
Uniform (Homogeneous)	Components are evenly distributed and cannot be seen separately even under a microscope.	Sugar solution, salt water, vinegar (acetic acid in water), aerated water, clean air, alloys.
Alloys (Solid-Solid)	Uniform mixtures of metals or metals with non-metals that retain combined strength and properties.	Stainless steel (Fe, Ni, Cr, C), Brass (Cu + Zn), Bronze (Cu + Sn).

Air as a Uniform Mixture: Air consists of nitrogen (~78%), oxygen, argon, carbon dioxide, and water vapour. Oxygen supports life and combustion, whereas nitrogen does not take part in combustion. Air also contains varying particulate pollutants (dust, soot, CO, NO₂, SO₂).

Key Activity & Heritage Highlight

Limewater Test for CO₂ (Activity 8.1): Exposing limewater (calcium hydroxide) to carbon dioxide in air turns it *milky* due to insoluble calcium carbonate:

Calcium Hydroxide + Carbon Dioxide → Calcium Carbonate (insoluble white ppt) + Water

Indian Scientific Heritage: Ancient texts like *Charaka Samhita* mention *Mishraloha* (alloys) used for medicinal and practical purposes, such as *Kamsya* (Bronze: 4 parts Copper + 1 part Tin).

3. Elements — Building Blocks of Matter

An **element** is a pure substance made up of identical particles called *atoms* that cannot be broken down into simpler substances by any chemical or physical reaction. Elements are the fundamental building blocks of all matter. Currently, **118 elements** are known.

- **States of Elements:** Most elements are solids at room temperature. 11 elements exist as gases (all non-metals, e.g., O₂, N₂, He). Only 2 elements are liquids at room temperature: **Mercury** (metal) and **Bromine** (non-metal). Gallium and Caesium turn liquid at ~30 °C.
- **Classification:** Elements are categorized into *Metals* (gold, iron, copper), *Non-metals* (oxygen, carbon, sulfur), and *Metalloids* (silicon, boron) displaying intermediate properties.
- **Molecules of Elements:** Atoms of most elements combine to form stable independent particles called *molecules* (e.g., two hydrogen atoms form one H₂ molecule; two oxygen atoms form O₂).

4. Compounds — Chemical Combinations

A **compound** is a pure substance formed when two or more different elements combine chemically in **fixed ratios**. The properties of a compound are entirely distinct from its constituent elements, and its elements cannot be separated by physical methods.

Key Experiment / Activity	Observation & Reaction	Scientific Conclusion
Electrolysis of Water (Activity 8.3)	Passing electricity through acidified water yields Hydrogen gas (pop sound) and Oxygen gas (flame glows brighter) in a 2:1 volume ratio.	Water (H_2O) is a compound made of Hydrogen and Oxygen chemically combined in a fixed 2:1 atomic ratio.
Heating Sugar (Activity 8.4)	Gently heating sugar turns it brown then chars it to black charcoal (Carbon) while condensing water droplets inside the tube.	Sugar is a compound composed of Carbon, Hydrogen, and Oxygen combined chemically.
Iron + Sulfur Comparison (Activity 8.5)	Sample A (Unheated physical mix): Iron filings pulled by magnet; reacts with HCl releasing H_2 gas (pop sound). Sample B (Heated to black mass): Non-magnetic; reacts with HCl releasing H_2S gas (rotten egg odor).	Sample A is a <i>mixture</i> (retains properties). Sample B is a <i>compound</i> (Iron Sulfide, FeS) with brand new properties.

5. Key Differences: Elements, Compounds & Mixtures

Property	Element	Compound	Mixture
Composition	Made of only one type of atom.	Two or more elements chemically combined in fixed ratios.	Two or more substances physically mixed in any proportion.
Properties	Displays characteristic elemental properties.	Properties are completely different from constituent elements.	Components retain their individual original properties.
Separation	Cannot be broken down simpler by physical or chemical means.	Constituent elements cannot be separated by physical methods.	Components can be separated easily by physical methods.
Chemical Reaction	Fundamental building block.	Formed as a result of a chemical reaction.	No chemical reaction occurs during formation.
Examples	Iron (Fe), Oxygen (O_2), Gold (Au), Carbon (C).	Water (H_2O), Salt (NaCl), Sugar, Iron Sulfide (FeS).	Air, Sprout salad, Seawater, Brass, Stainless steel.

6. Minerals, Modern Materials & Traditional Art

- **Minerals:** Natural solid substances in Earth's crust with fixed chemical compositions. Most are compounds (quartz, calcite, mica, talc, pyroxene, olivine), while some are *native elements* (gold, silver, copper, sulfur).
- **Graphene Aerogel:** An advanced ultralight carbon material ('wonder material') capable of resting on grass. Highly porous with high absorption capacity, making it ideal for environmental oil spill cleanups.
- **Dhokra Art:** Ancient wax-casting metal craft from Bihar & Odisha using molten brass or bronze (alloys) to create traditional figures.

Summary Snapshot / Exam Essentials

1. **Matter** has mass and volume. Light, heat, thoughts, and emotions are NOT matter.
2. **Mixtures** can be uniform (homogeneous) or non-uniform (heterogeneous); components retain properties and can be separated physically.
3. **Pure Substances** are either **Elements** (118 known, single type of atom) or **Compounds** (fixed ratio of different elements, new properties).
4. **Limewater + CO_2** → Calcium Carbonate (turns milky).
5. **Water Electrolysis:** 2 volumes Hydrogen (pop sound) : 1 volume Oxygen (glowing splint).
6. **Iron + Sulfur (Unheated):** Mixture (Fe magnetic). **Heated (FeS):** Compound (Fe non-magnetic, releases H_2S rotten egg gas with HCl).